

# Trajectory planning for the 🏭 UFactory Lite 6 Robot arm using Manufacturer's Python SDK

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## 1. Abstract

## 2. Introduction

This document aims to shed light on the workflow and intricacies involved in writing software used to for trajectory planning and job execution for the 🏭 *UFactory Robot arm*, model *Lite 6*. We will be looking at it from the context of the Big Chemistry project. In the following chapters, you will notice the different types of  DOT research methods that come into play for the duration of the project's task.

## 3. What is 🏭 UFactory Lite 6 Robot arm?

As the name suggests, the machine is a mechanized non-humanoid arm that is operated with manufacturer's own software or can be programatically set to execute instructions using, again, a Standard Development Kit from 🏭 UFactory(manufacturer). The program instructions are written in  Python and 🏭 UFactory provides a web interface that connect's to the arm's server, which includes a  Python IDE, and allows a developer to write logic for the robot, much like how one would write software for an Arduino, using their API.

## 4. Why the Lite 6?

The Lite 6 model is one of the best budget options 🏭 UFactory provides and its low-weight payload handling(up to 600 grams) is a great match for what the robot's function will be.

The job of the arm is to transport low-weight payload(a small spot plate) from an  OpenTrons Pippeting Robot, to a Tecan Spark plate reader. The chemists that will use this setup for their daily lab activites must have reliable equipment that allows them to handle the substances they will research with great care.

**5. How the  Python SDK is leveraged? ( Library research)**

**6. Conclusion**

**7. Appendix**

**7.1. Sources**